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Applied Analytics Project - Week 3 Assignment

Dataset Partition Strategy:

The dataset was partitioned using a chronological train-test split rather than a random split. Specifically, March Madness games from 2003-2019 were used for the training set, while games from 2021-2024 were used for the testing set. The 2020 tournament was cancelled due to COVID, so there is no data for that year.

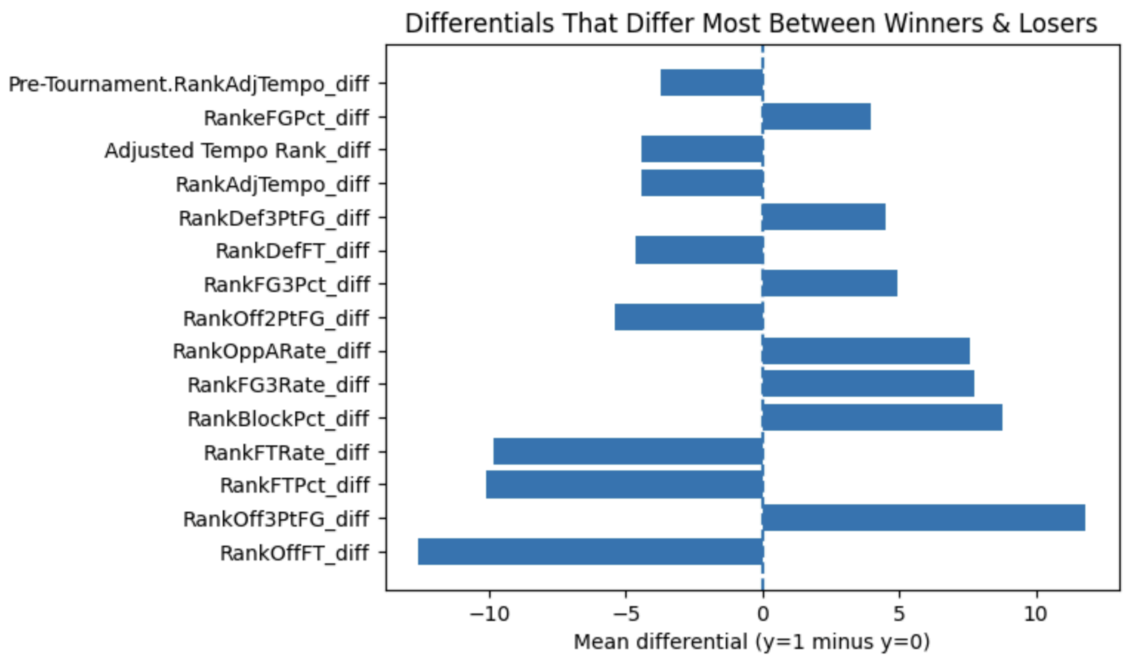
This strategy was used to prevent data leakage and attempt to mirror the way a real life predictive model would be used. Tournaments occur every year, so allowing information from future seasons to influence model training has the potential to inflate performance. This type of split ensures that the model is trained only on historical information when making predictions on unseen tournaments.

Using multiple recent tournaments as the testing set also creates a more stable assessment of model performance. March Madness is inherently volatile which means evaluating the model on a single season could lead to poor conclusions caused by randomness and season specific variability.

As the 2026 March Madness tournament begins in a little over a month, new matchups will present themselves. This presents the opportunity to test the models on real time data, and potentially explore the model outcomes in betting markets.

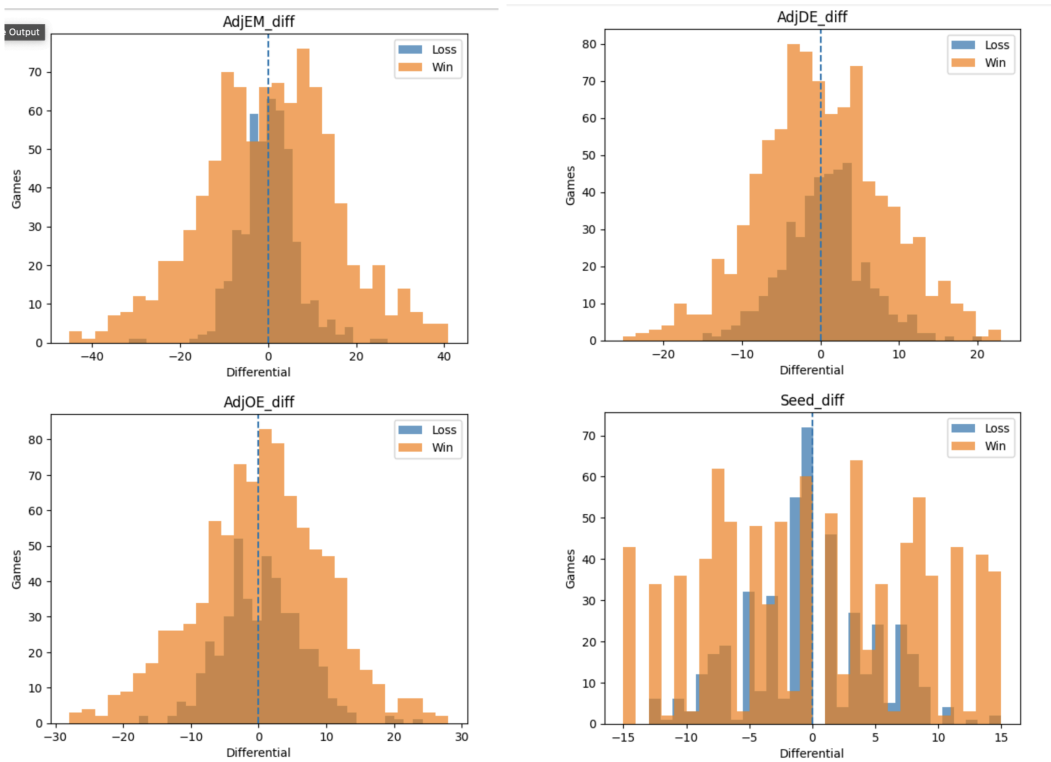
EDA Results:

Highest Differentials Between Winners and Losers



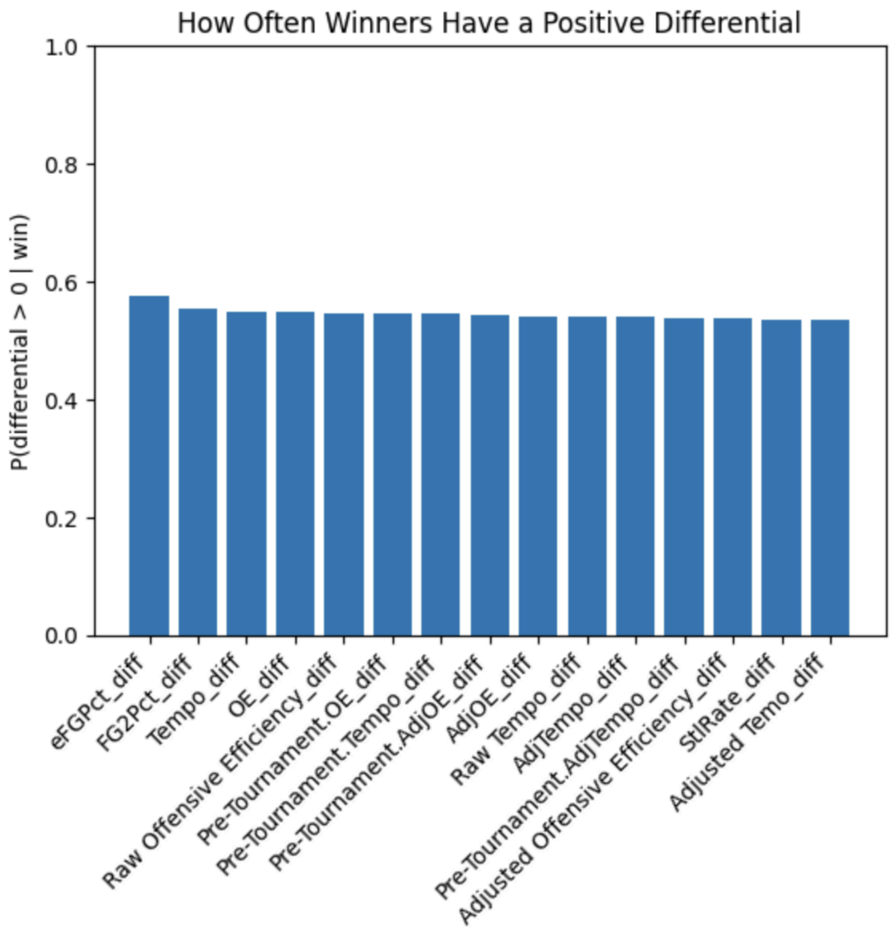
This figure shows which stat differentials separate winning and losing teams the most on average, helping identify the strongest potential predictors of game outcomes.

Distributions of Winners and Losers for Key Statistics



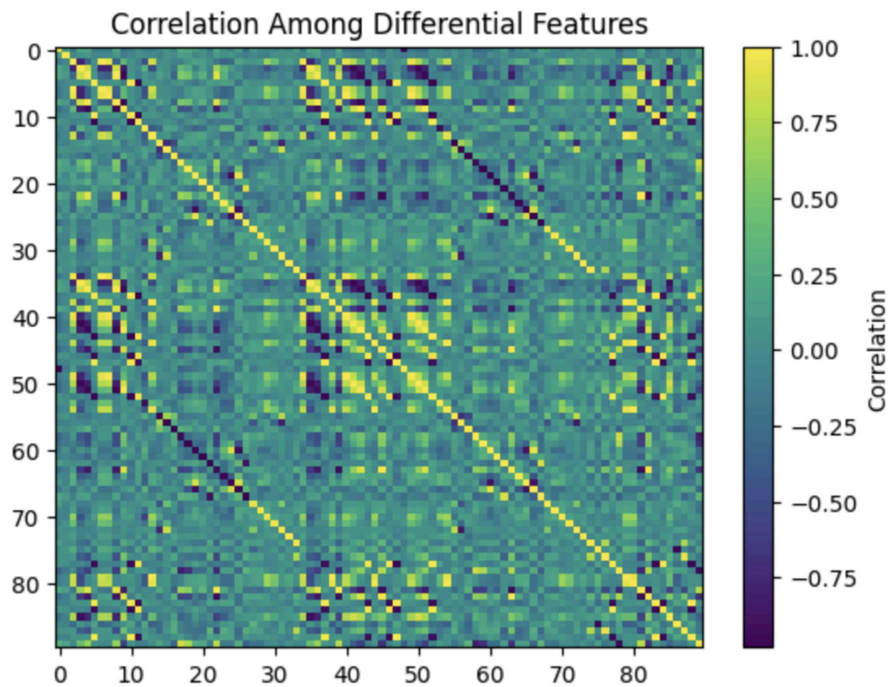
These histograms compare how key stat differentials are distributed across wins and losses which shows the separation between the two groups.

<u>Percentage of Winners with a Positive Differential</u>



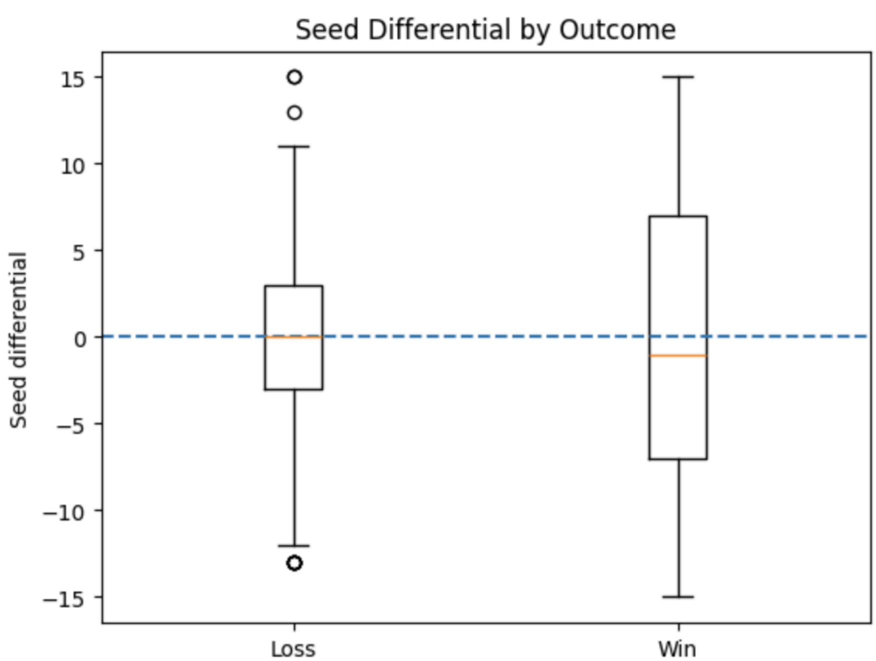
This plot shows how often the winning team had the advantage in each statistic.

<u>Correlation Heatmap for Differential Features</u>



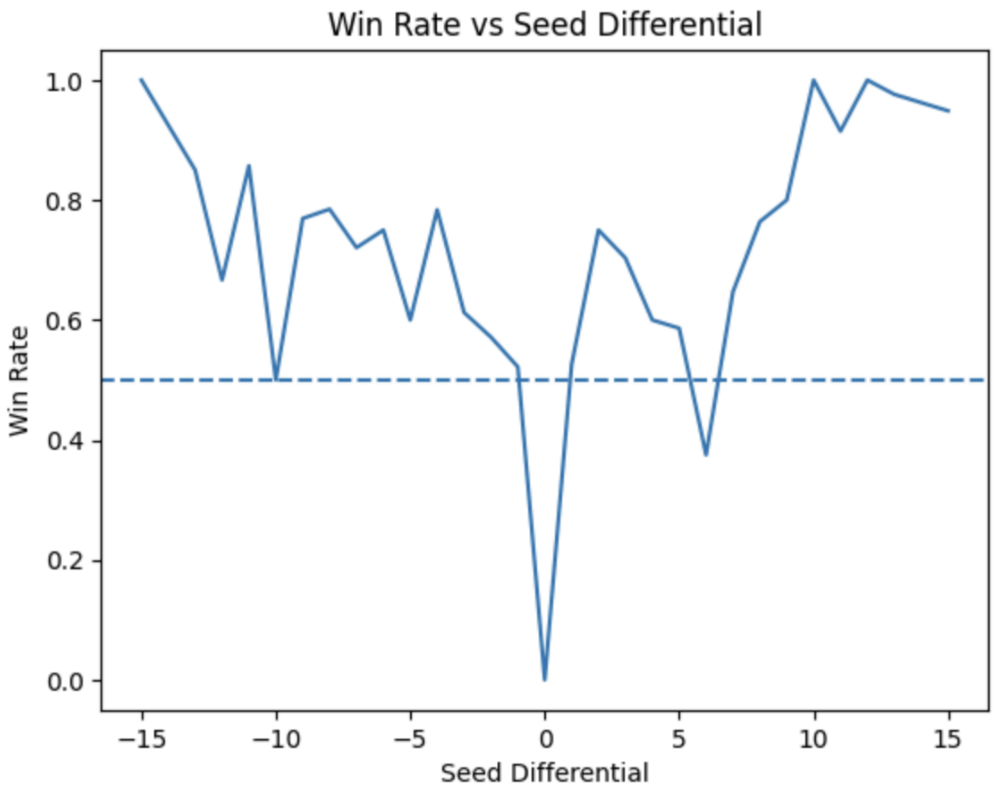
The heatmap shows the relationships between the variables to help identify clusters of related features to lower redundancy before modeling, although it is difficult to read clearly due to the large number of variables shown at once.

<u>Seed Differential by Outcome</u>



This visualization compares seed gaps in wins versus losses which highlights how seeding differences can relate to game outcomes.

<u>Win Rate vs Seed Differential</u>



This figure shows how the probability of winning changes as the seed gap between teams increases.

EDA Insights:

The EDA results show that the dataset contains some signals that prove helpful in predicting March Madness winners/losers, but there is still a lot of overlap in statistics between some of the winning and losing teams. Metrics such as season shooting efficiency and free-throw percentage were some of the most important signals that could predict a winner of a game. This isn't very shocking, as scoring efficiently is possibly the most important thing in the sport. It is important to note that there were still plenty of instances where a sharpshooting team lost, showing how volatile the tournament is.

Other indicators, including tournament seeding, also tended to show a clear outcome, with the win rates increasing as the difference between seeds grows. The EDA results show that the dataset has plenty of strong signals to generate solid win probabilities and identify any potential upsets. A major challenge that arose from the EDA is that many of the matchups look similar, with some of the strongest indicators. This can be a problem since many games will result in the model not being entirely confident in its output.

Data Problems:

One issue we identified is that the dataset is built from a team1 vs. team2 perspective which can make some features like seed differential harder to interpret since a positive or negative value does not always clearly represent the stronger team. For example, large positive and negative seed gaps (15 vs -15) both show high win rates which depend on whether the stronger team was assigned as team1 making our interpretations less consistent.

We also noticed from our EDA that games with a seed differential of 0 always resulted in a loss (72 occurrences). This tells us that team1 is consistently recorded as the losing team in matchups where both opponents are seeded equally. The heatmap also suggests some features may be highly correlated, which could create redundancy.

To resolve these issues, we plan to be more careful with how we interpret differential features so that positive values consistently reflect an advantage for team1. The amount of predictors we use will be reduced by removing variables that are highly correlated to one another. We'll also either adjust or remove the cases where the seed differential equals 0 to reduce its potential negative influence on our models.